

Synchronous Convergence and Applications

1. Preliminary definitions

Definition 1.1. A set I together with a relation $\leq$ is said to be *directed* by $\leq$ if the following properties hold:

- **Reflexivity:** $i \leq i$ for all $i \in I$;
- **Transitivity:** If $i_1 \leq i_2$ and $i_2 \leq i_3$, then $i_1 \leq i_3$;
- **Upper bound property:** For all $i_1, i_2 \in I$, there is $i \in I$ such that $i_1 \leq i$ and $i_2 \leq i$.

Definition 1.2. Let I, J be two directed sets. A map $\sigma : J \rightarrow I$ is said to be *directed* if for any $i_0 \in I$ there is $j_0 \in J$ such that $j \geq j_0$ implies $\sigma(j) \geq i_0$.

Definition 1.3. Let X be a set and let I be a directed set. A function $I \rightarrow X$ is called a *net* in X , denoted by $(x_i)_{i \in I}$. The set of all nets in X (with varying directed sets) will be denoted by $\mathcal{N}(X)$.

Definition 1.4. Let X be a topological space, and consider a net $(x_i)_{i \in I}$ in X . We say that $(x_i)_{i \in I}$ *converges* to $x \in X$ and write $x_i \xrightarrow{i \in I} x$, if for any neighborhood U of x , there is $i_0 \in I$ such that $i \geq i_0$ implies $x_i \in U$. In other words, x_i is *eventually* in any neighborhood of x .

Definition 1.5. Let $(x_i)_{i \in I}$ be a net in a set X . If $\sigma : J \rightarrow I$ is a directed map, then the net $(x_{\sigma(j)})_{j \in J}$ is called a *subnet* of $(x_i)_{i \in I}$.

Definition 1.6. Let X be a set together with a family $\mathcal{C} \subset \mathcal{N}(X) \times X$. Then $(X, \mathcal{C})$ is called a *convergence space* if the following conditions are satisfied:

- For any $x \in X$ and any directed set I , we have $((x_i)_{i \in I}, x) \in \mathcal{C}$. In other words, any constant net $\mathcal{C}$ -converges to the constant;
- If $((x_i)_{i \in I}, x_0) \in \mathcal{C}$, then $((x_{\sigma(j)})_{j \in J}, x_0) \in \mathcal{C}$ for any subnet $(x_{\sigma(j)})_{j \in J}$ of $(x_i)_{i \in I}$.

The condition $((x_i)_{i \in I}, x_0) \in \mathcal{C}$ may be denoted by $x_i \xrightarrow{i \in I}^{\mathcal{C}} x_0$ or just $x_i \xrightarrow{i \in I} x_0$.

Remark 1.1. Any topological space X automatically receives a convergence space structure as per [Definition 1.4](#).

Definition 1.7. A function $f : X \rightarrow Y$ between two convergence spaces is said to be *continuous* if for any net $(x_i)_{i \in I}$ converging to x in X , the net $(f(x_i))_{i \in I}$ converges to $f(x)$ in Y .

Definition 1.8. Let X be a set together with a family $\mathcal{C} \subset \mathcal{N}(X) \times X$. Then the family $\mathcal{C}$ is called a *convergence class* if the following hold:

- $(X, \mathcal{C})$ is a convergence space;
- Let $x \in X$. If every subnet of $(x_i)_{i \in I} \subset X$ has a subnet that $\mathcal{C}$ -converges to x , then $x_i \xrightarrow{i \in I}^{\mathcal{C}} x$.
- Consider a directed set I . For all $i \in I$, let J_i be also a directed set and consider a net $(x_j^i)_{j \in J_i}$ that $\mathcal{C}$ -converges to $x^i \in X$. Assume also that $x^i \xrightarrow{i \in I}^{\mathcal{C}} x \in X$. Denote

$F := I \times \prod_{i \in I} J_i$. The set J is directed by the pointwise order inherited from I and J_i 's. We then require that the following convergence must hold:

$$x_{f(i)}^i \xrightarrow[(i,f) \in F]{\mathcal{C}} x.$$

Theorem 1.1. (J. L. Kelley, see [1]): *Let X be a set with a convergence class $\mathcal{C}$. For a set $A \subset X$, define $\bar{A}$ to be the set of all $x \in X$ such that there is a net $(x_i)_{i \in I} \subset A$ that $\mathcal{C}$ -converges to x . Then map $A \mapsto \bar{A}$ is a closure operator that endows X with a topology in which a net $(x_i)_{i \in I} \subset X$ converges to $x \in X$ iff it $\mathcal{C}$ -converges to x .*

2. The topology of synchronous convergence

Theorem 2.1. *Let X and Y be two convergence spaces, and let $\mathcal{C} = \mathcal{C}(X, Y) \subset Y^X$ be the family of all continuous maps from X to Y . For a directed set I , a net $(f_i)_{i \in I} \subset \mathcal{C}$, and a point $f \in \mathcal{C}$, we say that f_i converges to f if and only if for every directed set J with a monotone final map $\sigma : J \rightarrow I$, and any net $(x_j)_{j \in J} \subset X$, we have*

$$f_{\sigma(j)}(x_j) \xrightarrow{j \in J} f(x) \in Y.$$

*We claim that the set $\mathcal{S}$ consisting of pairs $((f_i)_{i \in I}, f)$ such that $f_i \xrightarrow{i \in I} f$ as defined above, is a convergence class on $\mathcal{C}$, and subsequently provides a topology on $\mathcal{C}$, called the topology of **semi-uniform convergence**.*

Proof: We first need to verify four axioms that a convergence class must satisfy:

1. Let I be a directed set, and let $f \in \mathcal{C}$. Since f is continuous, for any directed set J and any net $(x_j)_{j \in J} \subset X$ that converges to $x \in X$, we have $f(x_j) \xrightarrow{j \in J} f(x) \in Y$.
Therefore the net $(f)_{i \in I}$ converges to f in $\mathcal{C}$.
2. We need to show that any subnet of a convergent net converges to the same point. Let $(f_i)_{i \in I} \subset \mathcal{C}$ be a net that converges to $f \in \mathcal{C}$, and let J be a directed set with $\sigma : J \rightarrow I$. To show that the subnet $(f_{\sigma(j)})_{j \in J}$ converges to f , consider another directed set K , mapped to J via $\tau : K \rightarrow J$, and a net $(x_k)_{k \in K} \subset X$ that converges to $x \in X$. Since the map $\sigma \circ \tau : K \rightarrow I$ is monotone and final, by the definition of convergence we see that

$$f_{\sigma(\tau(k))}(x_k) \xrightarrow{k \in K} f(x),$$

which means that $f_{\sigma(j)} \xrightarrow{j \in J} f$ in $\mathcal{C}$, as desired.

3. Let $(f_i)_{i \in I} \subset \mathcal{C}$ be a net and assume that every subnet of $(f_i)_{i \in I}$ has a subnet that converges to $f \in \mathcal{C}$. We need to show that the net $(f_i)_{i \in I}$ itself converges to f . To that end, let J be a directed set with a map $\sigma : J \rightarrow I$, together with a net $(x_j)_{j \in J} \subset X$ that converges to $x \in X$. Assume that the net $(f_{\sigma(j)}(x_j))_{j \in J}$ does **not** converge to $f(x)$ in Y . That means that there is a neighborhood U of $f(x)$ and a subnet $(f_{\sigma(\tau(k))}(x_{\tau(k)}))_{k \in K}$, indexed by a directed set K with $\tau : K \rightarrow J$, such that $f_{\sigma(\tau(k))}(x_{\tau(k)}) \notin U$ for all $k \in K$.
At the same time, defining $\alpha : K \rightarrow I$ by $\alpha = \sigma \circ \tau$, we see that $(f_{\alpha(k)})_{k \in K}$ is a subnet of $(f_i)_{i \in I}$. Therefore, this subnet itself has a subnet that converges to f , i.e. there is

a directed set L with a map $\beta : L \rightarrow K$, such that $f_{\alpha(\beta(l))} \xrightarrow{l \in L} f \in \mathcal{C}$. In particular, this implies that

$$f_{\sigma(\tau(\beta(l)))}(x_{\tau(\beta(l))}) \xrightarrow{l \in L} f(x).$$

But this is contradictory, since $f_{\sigma(\tau(\beta(l)))}(x_{\tau(\beta(l))}) \notin U$ for all $l \in L$.

4. Finally, we need to show the diagonal property. Let $(f_i)_{i \in I} \subset \mathcal{C}$ be a net converging to $f \in \mathcal{C}$. For every $i \in I$, consider also a directed set J_i and a net $(f'_{(i,j)})_{j \in J_i} \subset \mathcal{C}$ that converges to f_i . Let

$$K := I \times \prod_{i \in I} J_i, \quad I \times J := \{(i, j) \mid i \in I, j \in J_i\},$$

with the preorder on K defined pointwise, and consider $R : K \rightarrow I \times J$ defined by $R(i, \alpha) = (i, \alpha(i))$. We then need to show that the net

$$(f'_{R(i, \alpha)})_{(i, \alpha) \in K} \subset \mathcal{C}$$

converges to f . To that end, consider a directed set L with $\tau : L \rightarrow K$ and a net $(x_l)_{l \in L} \subset X$ converging to $x \in X$. We need to show the convergence

$$f'_{R(\tau(l))}(x_l) \xrightarrow{l \in L} f(x) \in Y.$$

Since the above convergence is topological, we can show it in the usual way. Let U be a neighborhood of $f(x)$ in Y . We seek $l_0 \in L$ such that

$$\forall l \geq l_0, f'_{R(\tau(l))}(x_l) \in U.$$

Consider the map $\eta : L \rightarrow I$ defined by $\eta(l) = i$, where $(i, \alpha) = \tau(l)$. Clearly, η is a monotone final map, so by the definition of convergence, $f_i \xrightarrow{i \in I} f$ implies that

$$f_{\eta(l)}(x_l) \xrightarrow{l \in L} f(x) \in Y.$$

Hence, there is $l_0 \in L$ such that for all $l \geq l_0$, we have $f_{\eta(l)}(x_l) \in U$. Now, for any l , we have the convergence

$$f'_{(\eta(l), j)} \xrightarrow{j \in J_{\eta(l)}} f_{\eta(l)} \in \mathcal{C}.$$

Applying the definition of convergence for the constant net $(x_l)_{j \in J_{\eta(l)}} \subset X$, we see that

$$f'_{(\eta(l), j)}(x_l) \xrightarrow{j \in J_{\eta(l)}} f_{\eta(l)}(x_l) \in Y.$$

Therefore, for all $l \geq l_0$, there is $j_0^l \in J_{\eta(l)}$ such that for all $j \geq j_0^l$, we have $f'_{(\eta(l), j)}(x_l) \in U$. Next, we define $\alpha_0 \in \prod_{i \in I} J_i$ as follows:

$$\alpha_0(i) := \begin{cases} j_0^l, & \text{if } i = \eta(l) \text{ with } l \geq l_0, \\ \text{any } j \in J_i, & \text{otherwise.} \end{cases}$$

We then have $(\eta(l_0), \alpha_0) \in K$, and since $\tau : L \rightarrow K$ is final, there is $l_1 \in L$ such that $l_1 \geq l_0$ and $\tau(l_1) \geq (\eta(l_0), \alpha_0)$.

Now, consider any $l \geq l_1$, with $\tau(l) = (i, \alpha)$. Since $\tau(l) \geq \tau(l_1) \geq (\eta(l_0), \alpha_0)$, we see that $\alpha \geq \alpha_0$, and so $\alpha(i) \geq \alpha_0(i)$ for all $i \in I$. Since $l \geq l_0$ and $\alpha(\eta(l)) \geq \alpha_0(\eta(l)) = j_0^l$, we have

$$f'_{R(\tau(l))}(x_l) = f'_{(\eta(l), \alpha(\eta(l)))}(x_l) \in U.$$

Since the neighbourhood U was arbitrary, this implies the convergence

$$f'_{R(\tau(l))}(x_l) \xrightarrow{l \in L} f(x) \in Y,$$

as desired.

This shows that our notion of convergence on $\mathcal{C}$ is a convergence class, and hence the closure operator defined by

$$\overline{E} := \left\{ f \in \mathcal{C} \mid \exists (f_i)_{i \in I} \subset E \text{ st } f_i \xrightarrow{i \in I} f \right\}$$

endows $\mathcal{C}$ with a topology in which convergence coincides with the one defined by $\mathcal{S}$. ■

Proposition 2.1. *Let X and Y be topological spaces and assume that X is first countable. Construct the topology on $\mathcal{C} = \mathcal{C}(X, Y)$ as above. Then a sequence $\{f_k\}_{k=1}^\infty \subset \mathcal{C}$ will converge to $f \in \mathcal{C}$ if and only if $f_k(x_k) \rightarrow f(x)$ in Y whenever $x_k \rightarrow x$ in X .*

Proof: Consider a sequence $\{f_k\}_{k=1}^\infty \subset \mathcal{C}$. Suppose that it converges to a point $f \in \mathcal{C}$. Note that this sequence is a net with $I = \mathbb{N}$. Taking $J = I = \mathbb{N}$ and $\sigma = \text{id}$, we see that for any sequence $\{x_k\}_{k=1}^\infty \subset X$ converging to $x \in X$, we have

$$f_k(x_k) \xrightarrow{k \rightarrow \infty} f(x). \quad (1)$$

Conversely, suppose that $f_k(x_k) \rightarrow f(x)$ whenever $x_k \rightarrow x \in X$. Let us first see that for any subsequence $\{f_{k_l}\}_{l=1}^\infty$ and any sequence $x_l \rightarrow x \in X$, we have $f_{k_l}(x_l) \rightarrow f(x) \in Y$. Indeed, given such sequences, for any $k \in \mathbb{N}$ we define $\alpha(k)$ to be the minimum of all $l \in \mathbb{N}$ such that $k \leq k_l$. Then the sequence $\{x_{\alpha(k)}\}_{k=1}^\infty$ converges to x , because if $x_l \in V \ni x$ for all $l \geq l_0$, then for all $k \geq k_{l_0}$, we have $\alpha(k) \geq \alpha(k_{l_0}) = l_0$, and so $x_{\alpha(k)} \in V$. Now, since $\alpha(k_l) = l$ for all $l \in \mathbb{N}$, the sequence $\{f_{k_l}(x_l)\}_{l=1}^\infty$ is a subsequence of $\{f_k(x_{\alpha(k)})\}_{k=1}^\infty$. And since the latter converges to $f(x)$ by (1), so does the former.

Finally, we show that $f_k \xrightarrow{k \rightarrow \infty} f \in \mathcal{C}$. Let J be a directed set with a monotone final map $\sigma : J \rightarrow \mathbb{N}$, and let $(x_j)_{j \in J} \subset X$ be a net that converges to $x \in X$. We need to show that the net $(f_{\sigma(j)}(x_j))_{j \in J}$ converges to $f(x)$. Assume the contrary. Then we have a neighborhood U of $f(x)$ and a subnet $(f_{\sigma(\tau(l))}(x_{\tau(l)}))$, with $\tau : L \rightarrow J$, such that

$$\forall l \in L, f_{\sigma(\tau(l))}(x_{\tau(l)}) \notin U. \quad (2)$$

Since X is first countable, since the net $(x_{\tau(l)})_{l \in L}$ converges to x , we can find a sequence $\{l_i\}_{i=1}^\infty$ such that the sequence $\{x_{\tau(l_i)}\}_{i=1}^\infty$ converges to x . Furthermore, by ensuring that $\sigma(\tau(l_{i+1})) > \sigma(\tau(l_i))$, we can define $k_i := \sigma(\tau(l_i))$ and obtain a subsequence $\{f_{k_i}\}_{i=1}^\infty$ of $\{f_k\}_{k=1}^\infty$. As shown before, we then have

$$f_{\sigma(\tau(l_i))}(x_{\tau(l_i)}) \xrightarrow{i \rightarrow \infty} f(x),$$

| which contradicts (2). It then follows that $f_k \xrightarrow[k \rightarrow \infty]{} f$, as desired. ■

Proposition 2.2. *If X is a topological space and Y is a TVS, then the space $\mathcal{C}(X, Y)$, with the topology of semi-uniform convergence and pointwise linear structure, is also a TVS.*

| *Proof:* Follows easily from the definition of the topology on $\mathcal{C}(X, Y)$ and the fact that Y is a TVS. ■

Definition 2.1. Let X be a first countable topological vector space, and let $n \in \mathbb{N}_0$.

The *curried function set* $H_n(X)$ is defined as follows:

- If $n = 0$, we set $H_0(X) := X$;
- If $H_n(X)$ is defined, we define $H_{n+1}(X)$ to be the set of all functions from X to $H_n(X)$.

The *curried continuous space* $C_n(X)$ is a TVS, defined recursively:

- If $n = 0$, we set $C_0(X) := X$;
- If $C_n(X)$ is defined, we set $C_{n+1}(X) := \mathcal{C}(X, C_n(X))$, with pointwise linear structure and the topology of semi-uniform convergence.

Remark 2.1. Any function $f \in C_n(X)$ naturally corresponds to a multivariable function $\tilde{f} : X^n \rightarrow X$:

$$\tilde{f}(x_1, x_2, \dots, x_n) = f(x_1)(x_2) \dots (x_n).$$

Lemma 2.1. *For all $n \in \mathbb{N}$, a function $f \in H_n(X)$ lies in $C_n(X)$ if and only if the corresponding function $\tilde{f} : X^n \rightarrow X$ is continuous.*

| *Proof:* First assume that $f \in C_n(X)$. To show that $\tilde{f}$ is continuous, consider a sequence $x_k^{(n)} \rightarrow x_0^{(n)} = (x_0^1, x_0^2, \dots, x_0^n) \in X^n$. Since f is continuous, we see that $f(x_k^1) \rightarrow f(x_0^1)$. If $n = 1$, we are done. Otherwise, the convergence $f(x_k^1) \rightarrow f(x_0^1)$ occurs in the topology of semi-uniform convergence, and so by Proposition 2.1 we have

$$f(x_k^1)(x_k^2) \rightarrow f(x_0^1)(x_0^2).$$

Repeating this argument as needed, we obtain

$$f(x_k^1)(x_2^1) \dots (x_k^n) \rightarrow f(x_0^1)(x_0^2) \dots (x_0^n),$$

which is to say that $\tilde{f}(x_k^{(n)}) \rightarrow \tilde{f}(x_0^{(n)})$, as desired.

We will prove the converse by induction.

- If $n = 1$, then the statement is trivial.
- Suppose the statement is proven for $C_n(X)$. Consider $f \in H_{n+1}(X)$ such that $\tilde{f} : X^{n+1} \rightarrow X$ is continuous. By the induction hypothesis, we know that $f(x) \in C_n(X)$ for all $x \in X$, so f is a function from X to $C_n(X)$. It remains to show that f is continuous. This is done in the same way as the first part of our proof. ■

Bibliography

[1] J. L. Kelley, *General Topology*, vol. 27. Graduate Texts in Mathematics, 1975.